Assume that in Kisumu County, the distribution of blood types is Type O = 42%, Type A = 41%, Type B = 10%, and Type AB = 5%.

a) An individual is picked at random from Kisumu County, what the probability that they have blood type O? A? B? AB?

b) What is the probability that the individual will have **either** blood group O **or** A?

   i) We know that a person cannot be in two blood groups, so these outcomes are mutually exclusive.

   ii) If two outcomes are mutually exclusive, then the addition rule of probability states that the probability of either one or the other occurring is the sum of the two individual probabilities.

   iii) Hence the probability of being in either blood group O or blood group A is

$$0.42 + 0.41 = 0.83.$$

Referring to Figure 1, what is the probability of a patient chosen at random being either satisfied OR very satisfied with their nursing care?

The multiplication rule

The multiplication rule states

$$p(\text{A and B}) = p(\text{A}) \times p(\text{B}|\text{A}) \tag{3}$$

If two outcomes A and B are independent then

$$p(\text{A and B}) = p(\text{A}) \times p(\text{B}) \tag{4}$$

Equation 3 gives the multiplication rule of probability.

Suppose you are interpreting blood type in a given county. It is known that 20% of the time a patient has blood type O in this county.

- The probability that you test one's blood and it is type O is 20%.
- The probability that a successive test is blood type O is 20%.
- This is because the results are independent of each other.

– When events are independent an outcome in no way influences the result of the next outcome.

What is the probability that two successive outcomes will both be blood type O?
Let the event that you test blood type O be A. Then

$$p(\text{A and A}) = p(\text{A}) \times p(\text{A}) = 0.2 \times 0.2 = 0.04$$

This means that if we repeat this experiment (two blood type interpretations in succession) a hundred times, on four occasions we would get two successive blood type O outcomes.

Conditional probability

Conditional probability is concerned with the probability of one outcome occurring given that another outcome has already occurred i.e. $p(A|B)$ - read probability that event A occurs given that B has occurred.

– For example, given that a baby has a birthweight less than 1500 g, what is the probability that its mother smoked while pregnant?

Conditional probabilities are calculated as:

$$p(A|B) = \frac{p(\text{A and B})}{p(B)} \tag{5}$$

A main application is in diagnostic testing and screening programs.

– What is the probability of having a disease given a positive screening test?

Conditional probability Example

Consider the following results for breast cancer screening in a county. A total of 64810 persons were screened. The result of the test was positive/negative. The persons tested either developed or did not develop breast cancer.

	Develop Breast Cancer	Breast Cancer Free	Total screened
Positive	132	983	1115
Negative	45	63650	63695
Total cancer	177	64633	64810

What is the probability of having the disease (A) given a positive screening (B) result? i.e. $p(A|B)$

- 1115 have positive screen and 132 of those have cancer; hence probability is $\frac{132}{1115}$.
- Using Equation(5),
  - $p(\text{cancer and positive}) = \frac{132}{64810}$
  - $p(\text{positive}) = \frac{1115}{64810}$
  - $p(A|B) = \frac{132}{64810} / \frac{1115}{64810} = \frac{132}{1115}$

Class Exercise

1. In a Local Authority (LA) area with a population of 100,000, there are 25,000 smokers, 875 of whom have experienced (non-fatal) ischaemic heart disease (IHD) in the past 5 years. Of the non-smokers, 750 have also suffered IHD in the same period. If a person is chosen at random from this LA, what is the probability that

   a) this person is a smoker?

   b) has had (non-fatal) IHD in the last 5 years?

   c) is a smoker and has suffered (non-fatal) IHD?

- The probability that a smoker is chosen is $\frac{\text{Number of smokers in population}}{\text{Total population}} = \frac{25000}{100000} = 0.25$
- The probability the person has had IHD is $\frac{875 + 750}{100000} = 0.01625$
- The probability that the person chosen is a smoker who has suffered IHD is $\frac{875}{25000} = 0.035$

Introduction

- things vary.

GREAT LAKES UNIVERSITY
UNIVERSITY OF KISUMU
COURSE OUTLINE

SCHOOL:	Community Health & Development
DEPARTMENT:	
CAMPUS/SEMESTER:	Kisumu Campus/January-April 2026
COURSE CODE/TITLE:	: BIOSTATISTICS
COURSE LECTURER:	Christine Akinyi Odieny **SIGNATURE**:
CONTACT:	**odienychristine42@outlook.com**

COURSE DESCRIPTION: Basic statistical measurements of central tendency and dispersion, means, median, hypothesis testing, estimation, confidence intervals, correlation and regression analysis; Probability and Inference. Theoretical Distributions: The principles, assumptions, on Normal, Binomial and Poisson distributions. Sampling techniques: Determination of sample size: Sample size estimation for various types of studies; case-control, evaluative, cross-sectional, prospective, clinical trials studies. Hypothesis Testing: Null Hypothesis, Alternative Hypothesis: Level of confidence; Type I and Type II errors; Power of the Test; Confidence intervals; Testing the difference between two sample means and proportions; Chi-Square and Student t-test, McNamar's test. Regression: Multiple regressions, logistic regression, fitting of model to data, confidence intervals for regression line, interpretation of regression outputs, meaning of R^2 Pearson coefficient. Analysis of variance: (ANOVA) Two-by-two tables, multi tables, assumptions, analysis, and interpretation of ANOVA tests.

ELOs: By the end of the course the learner will:

i) Enhance understanding of statistical measures and inference through practical application and analysis.
ii) Acquire a comprehensive knowledge of theoretical distributions through the study and application of their properties, characteristics, and statistical inference techniques.
iii) Develop proficiency in sampling techniques and the determination of sample size, demonstrated through the

application of appropriate sampling methods and accurate sample size calculations in practical research scenarios.

iv) Attain competence in hypothesis testing, regression analysis, and analysis of variance (ANOVA) through the successful application of these statistical methods in data analysis, interpretation of results, and drawing meaningful conclusions based on statistical inference.

Week	Topic	ELOs	Sub-Topic
1	Introduction to Biostatistics and Data Description	1. Describe the basic principles of biostatistics and its importance in healthcare research. 2. Apply appropriate measures of central tendency and dispersion to summarize and interpret biomedical data. 3. Describe different types of data and variables encountered in biostatistical analysis.	• Types of data and variables • Descriptive statistics: measures of central tendency and dispersion • Calculation and interpretation of means, medians, and measures of variability
2	Probability and Theoretical Distributions	1. Apply probability concepts and rules to calculate probabilities in the context of healthcare data. 2. Understand the principles and assumptions underlying the Normal, Binomial, and Poisson distributions in biostatistics. 3. Apply the properties of these distributions to model and analyze healthcare data.	• Basic concepts of probability theory • Probability rules and calculations • Probability distributions: Binomial, Poisson, Normal, t-, F- and Chi-square
3	Sampling and Sampling Techniques	1. Differentiate between Probability and Non-probability sampling approaches, understand their advantages and disadvantages, and apply appropriate sampling techniques based on the research context. 2. Explain the concept of sampling distributions and their role in statistical inference. 3. Apply the central limit theorem to estimate population parameters and make inferences about healthcare datasets.	• Probability and Non-Probability Sampling Techniques • Sampling distributions and the central limit theorem
4	Confidence Intervals	1. Construct and interpret confidence intervals for means, proportions, and other parameters in biostatistics. 2. Determine appropriate sample sizes for different types of studies in healthcare research. 3. Evaluate the precision and reliability of estimates using confidence intervals in the context of clinical trials and epidemiological studies.	• Construction and interpretation of confidence intervals • Confidence intervals for means and proportions • Sample size determination for different types of studies o Case-control, evaluative, cross-sectional, prospective, and clinical trials studies
5	Hypothesis Testing	1. Formulate null and alternative hypotheses for research questions in biostatistics. 2. Calculate and interpret p-values and make decisions based on hypothesis tests in the context of healthcare studies. 3. Understand the concepts of Type I and Type II errors and their implications in hypothesis testing in biostatistics.	• Null and Alternative hypotheses • Type I and Type II errors • Significance level, p-value, and critical regions • One-sample and two-sample hypothesis tests o Student's t-test and its applications • Chi-Square test for independence
6	ANOVA	1. Understand the concept and assumptions of ANOVA and its applications in healthcare research.	• Introduction to ANOVA • One-way ANOVA: assumptions, calculations, and interpretation.

		2. Perform one-way ANOVA to compare means of multiple groups in biomedical studies. 3. Interpret ANOVA results, including F-statistics and p-values, and conduct post-hoc tests for multiple comparisons.	• Two-way ANOVA and multi-way ANOVA. • Post-hoc tests and multiple comparisons
7	Non-Parametric Tests	1. Identify situations where non-parametric tests are applicable in biostatistical analysis. 2. Apply non-parametric tests such as the Wilcoxon signed-rank test, Mann-Whitney U test, and Kruskal-Wallis test to healthcare datasets. 3. Interpret non-parametric test results and understand their advantages and limitations compared to parametric tests.	• Introduction to non-parametric tests o Wilcoxon signed-rank test o Mann-Whitney U test o Kruskal-Wallis test o McNemar test
8	Correlation and Regression Analysis	1. Analyze and interpret the strength and direction of relationships between variables in healthcare datasets using correlation coefficients. 2. Apply simple linear regression analysis to model and predict health outcomes based on explanatory variables. 3. Interpret regression outputs, including coefficients, p-values, and R-squared, in the context of biomedical research.	• Understanding the relationship between variables • Calculation and interpretation of correlation coefficients • Simple linear regression analysis • Fitting a regression model to data • Confidence intervals and hypothesis testing for regression coefficients • Introduction to logistic regression
9	Advanced Topics in Statistical Inference	1. Understand the concept of statistical power and its importance in designing and interpreting healthcare studies. 2. Determine sample sizes for hypothesis testing and power calculations in biostatistics. 3. Apply meta-analysis techniques to combine and analyze results from multiple studies in healthcare research.	• Power of the test • Sample size determination for hypothesis testing • Non-inferiority and equivalence testing • Meta-analysis: combining results from multiple studies
10	Review and Applications	1. Review and consolidate key concepts and techniques covered throughout the course. 2. Apply statistical methods and reasoning to real-world scenarios in biostatistics. 3. Demonstrate critical thinking and ethical considerations in statistical analysis and data interpretation in the healthcare field.	• Recap of key concepts and techniques covered in the course • Application of statistical methods to real-world scenarios • Ethical considerations in statistical analysis

ASSESSMENT STRATEGY: Each week will consist of lectures, practical examples, and problem-solving sessions.
Assignments and quizzes will be given to reinforce understanding and assess learning progress.
End of Semester Examination

REFERENCES:

1. Bernard Rosner (2016) Fundamentals of Biostatistics 8th Edition. CENGAGE Learning. ISBN: 978-1-305-26892-0
2. Chang, Mark. (2013). Modern Issues and Methods in Biostatistics (Statistics for Biology and Health).

COD: ____________________________ **Signature:** ________________ **Date:** 30/1/2026

Module 11: Clinical Epidemiology & Diagnostic Test Statistics

Module Assignment

The supplied PDF contains conditional-probability and screening examples, including a breast-cancer screening table. It does not provide a dedicated full diagnostic-test chapter; the assignment extends the supplied screening material to the requested syllabus.

1. Using a simulated 2×2 diagnostic-test table, identify TP, FP, TN and FN.

2. Calculate sensitivity, specificity, positive predictive value and negative predictive value.

3. Calculate accuracy and positive and negative likelihood ratios.

4. Explain how changing disease prevalence can change PPV and NPV even when sensitivity and specificity remain unchanged.

5. Write a short clinical recommendation on whether the test is more suitable for screening or confirmation.

Submission guidance: Show working where calculations are required. Interpret statistical results in clinical language. Use Excel, Jamovi or SPSS where appropriate.